

Bayesian Optimization for Dose Finding with Two Agents: Participant Allocation and Final Selection

Xinzhu Wang Tanzy Love

Department of Biostatistics and Computational Biology, University of Rochester
xinzhu_wang@urmc.rochester.edu; tanzy_love@urmc.rochester.edu

Abstract

In two-agent dose-finding trials, the next cohort should help identify a combination for final selection. We studied a constrained knowledge-gradient (cKG) rule with one-cohort lookahead that updates independent Gaussian-process models of efficacy and continuous toxicity, reapplies a probability criterion for mean toxicity, and evaluates the resulting selection. We derived a deterministic calculation over a fixed set of dose combinations, holding fitted model parameters fixed during each hypothetical update. We compared cKG with constrained expected improvement (cEI) and two toxicity-only rules, targeted mean squared error (tMSE) and entropy, in four synthetic scenarios. In the primary obstructive sleep apnea (OSA)-derived scenario, averaged equally over strata and five probability cutoffs, cKG assigned fewer participants to combinations above the true mean-toxicity limit than tMSE (17.92% versus 27.08%), but selected such combinations more often at trial completion (18.80% versus 11.85%). Compared with cEI, cKG had higher mean simulated reduction in the 4%-desaturation apnea-hypopnea index (AHI4) at final selection (7.46 versus 6.72 events/hour), more above-limit final selections (18.80% versus 10.50%), and more above-limit assignments (17.92% versus 15.10%). Across scenarios, its efficacy advantage over cEI was smaller under stricter toxicity criteria. Continuous outcomes, uncalibrated toxicity limits, and a rule that still selects a combination when none meets the criterion limit clinical interpretation. Allocation and final-selection toxicity should be reported separately, alongside efficacy.

Keywords: Bayesian optimization; drug-combination dose finding; knowledge gradient; Gaussian process; efficacy and toxicity; operating characteristics.

1 Introduction

Dose finding with two agents requires a choice among combinations that may differ in both efficacy and toxicity. An optimal biological dose (OBD) is often defined through a benefit–risk criterion rather than toxicity alone^{1;2}. We maximize mean efficacy subject to an upper limit on mean toxicity. Two-agent dose combinations do not form a single ordered sequence: increasing one agent while decreasing the other need not reduce toxicity, and efficacy may plateau or decline. With a small trial and many dose pairs, most combinations receive little or no direct observation^{3;4}. We ask which eligible combination should receive the next cohort to inform the dose selected at trial completion.

Bayesian optimization (BO) combines a response model with a rule for choosing the next dose combination. Gaussian-process (GP) models use an assumed covariance structure to predict responses at other combinations from the observed outcomes. The allocation rule, or acquisition function, scores eligible combinations according to predicted efficacy, uncertainty about toxicity, or the expected effect of new observations on the final choice.

Existing dose-optimization designs address a broader set of decisions. BOIN12 and Comb-BOIN12 use utility-based, model-assisted rules; DROID identifies a therapeutic dose range before randomized comparison; and BARD combines backfill with adaptive randomization⁵⁻⁸. Penalized D-optimality has also been used to learn a combination-toxicity surface while respecting admissibility and escalation rules⁹. We hold the response models, assignment restrictions, and final-selection rule fixed to compare the objectives used to allocate the next cohort.

BO and information-directed allocation already have dose-finding applications^{4;10;11}. The combination-dose designs of Willard et al. use constrained expected improvement (cEI), weighting potential efficacy improvement by the probability of meeting a toxicity limit^{12;13}. Their updated design uses fully Bayesian GP fitting, dose escalation, and stopping based on uncertainty about the optimal dose¹³. We adapt the knowledge-gradient (KG) approach¹⁴⁻¹⁷ to ask how one additional cohort would change the best posterior mean efficacy among combinations meeting a toxicity criterion. Both response models are updated before that criterion is reapplied. The resulting constrained KG (cKG) score is positive when the expected value of the final selection increases, and can be negative when toxicity learning removes promising combinations from consideration. We derive a deterministic evaluator for a finite grid with independent Gaussian response models for efficacy and toxicity. We compare cKG with cEI on the same OSA-derived surface used by Willard et al. and on three additional surfaces. Two toxicity-only acquisitions, targeted mean squared error (tMSE) and an entropy-reduction rule, provide further comparators.

Our cKG formulation differs from existing constrained KG methods in its final-selection rule. Ungredda and Branke maximize posterior efficacy weighted by feasibility probability, corresponding to a specified penalty for an infeasible terminal choice¹⁷. Chen et al. combine expected improvement in the best constraint-mean-feasible prediction with query-feasibility weighting¹⁶. Here, a probability cutoff defines the selection set; if it is empty, a standardized-toxicity fallback determines the choice. The cKG score is the expected change in the value of this rule.

Obstructive sleep apnea (OSA) illustrates the trade-off between efficacy and tolerability. In MARIPOSA, placebo-adjusted changes in the 4%-desaturation apnea-hypopnea index (AHI4) were similar for the 2.5/75- and 5/75-mg aroxybutynin/atomoxetine groups: -7.16 and -7.20 events/hour. Dry mouth was reported in 24% and 59%, respectively. The later 26-week SynAIRgy trial evaluated the 2.5/75-mg pair^{18;19}. These fixed-arm trials motivate the dose-choice problem, although their data do not identify a two-dimensional dose-response surface. Our primary scenario uses the synthetic OSA model of Willard et al.¹². We also use an author-specified MARIPOSA-motivated surface, informed qualitatively by the published results, and generic logistic and Gaussian-bump surfaces. The logistic scenario uses a monotone dose-toxicity geometry familiar from early-phase dose-finding evaluations²⁰. In this scenario, both mean efficacy and mean toxicity increase with dose, restricting acceptable combinations to the lower-dose region. It tests the allocation and final-selection comparisons under that geometry, using synthetic continuous responses without calibration to a specific clinical program.

Our contribution is a deterministic calculation of the one-cohort cKG score and a comparison of allocation objectives under shared decision rules. The accompanying Python package, `dose-combination-bo`, implements all four acquisitions, with examples for running a trial and comparing rules. The archived study implementation remains the source of the reported operating characteristics.

2 Methods

2.1 Dose-selection target and response models

Let $e(d, z)$ and $g(d, z)$ be mean efficacy and mean toxicity for dose pair $d \in [0, 1]^2$ in stratum $z \in \{0, 1\}$. Both are continuous functions of dose; larger efficacy and smaller toxicity are preferred. The upper limit on mean toxicity is $g^\dagger(z)$. The toxicity score, observation window, and limit are specified simulation inputs. Assignment and final selection use the 5×5 grid $\mathcal{G} = \{0, 0.25, 0.50, 0.75, 1\}^2$. The coordinates are standardized dose levels: zero denotes the lowest dose level, which need not be placebo or absence of an agent. Every scenario contains at least one grid combination below the mean-toxicity limit. The target is

$$d_{\text{grid}}^*(z) = \arg \max_{d \in \mathcal{G}} e(d, z) \quad \text{subject to} \quad g(d, z) \leq g^\dagger(z). \quad (1)$$

This target maximizes efficacy within a toxicity constraint; it does not assign a further penalty to toxicity below that limit or require a minimum efficacy. The continuous surfaces e and g define simulation truth. GP covariance is also defined over the continuous domain, but assignment and final selection are restricted to $\mathcal{G}$. We fit independent Matérn-5/2 GPs to efficacy and toxicity. Write E and G for their latent random functions, with posterior means μ_e, μ_g and latent posterior SDs σ_e, σ_g . Kernel and mean parameters are re-estimated after each observed cohort. Let D_t contain the observed doses and both outcomes through cohort t , including initialization data. For the model and acquisition definitions below, fix a stratum and the current data D_t ; stratum and time arguments are suppressed unless needed. Let Φ be the standard normal distribution function. For $0 < \tau < 1$, define

$$z_g(d) = \frac{g^\dagger - \mu_g(d)}{\sigma_g(d)}, \quad \text{PF}(d) = \Pr\{G(d) \leq g^\dagger \mid D_t\} = \Phi\{z_g(d)\}, \quad (2)$$

$$A_\tau = \{d \in \mathcal{G} : \text{PF}(d) > \tau\}.$$

A_τ is the subset of the full dose grid $\mathcal{G}$ that passes the current posterior criterion. The probability $\text{PF}(d)$ concerns the unknown *mean* toxicity, not an individual participant's probability of an adverse event. The cutoff τ is therefore not a target dose-limiting-toxicity rate. For a fixed posterior, increasing τ makes the requirement more stringent. All rules use the strict inequality in Equation (2); cEI also uses $\text{PF}(d)$ as a continuous weight.

At the scheduled final analysis, the rule selects $\hat{d} = \arg \max_{d \in A_\tau} \mu_e(d)$ if any combination meets the criterion. Otherwise, it selects the grid combination with the largest z_g . This *fallback selection does not meet the posterior toxicity criterion*. No minimum-exposure requirement is imposed, so the selected combination may never have been assigned. Under gradual availability, a trial stopped by the rule below ends without recommending a combination.

2.2 Acquisition functions

We compare four ways to score the next cohort's assignment. cEI and cKG use efficacy and toxicity; tMSE and the entropy rule learn only about toxicity. All four are compared under the same assignment, stopping, and final-selection rules; tMSE and entropy serve as acquisition comparators rather than established trial designs. Let $\mathcal{G}_t \subseteq \mathcal{G}$ be the combinations currently available, and let $\mathcal{E}_t \subseteq \mathcal{G}_t$ be those permitted for the next assignment. Each acquisition ranks $A_\tau \cap \mathcal{E}_t$ when that set is nonempty. With only one eligible combination, the acquisition cannot change the assignment. If no permitted combination meets the criterion, the same highest- z_g fallback is applied within $\mathcal{E}_t$, subject to the stopping rule below. Ties are resolved by a fixed grid order.

For final-selection value, let $C_t = A_\tau \cap \mathcal{G}_t$ when nonempty; otherwise, C_t is the singleton containing the available combination with the largest z_g . Write $V_t = \max_{x \in C_t} \mu_e(x)$ for its best posterior mean efficacy. This value is also the cEI incumbent when the acquisition determines assignment.

Feasibility-weighted expected improvement (cEI). We rank dose combinations by

$$\alpha_{\text{cEI}}(d) = \text{EI}\{\mu_e(d), \sigma_e(d); V_t\} \text{PF}(d). \quad (3)$$

EI is the posterior expected positive efficacy gain over V_t . The incumbent includes previously visited combinations and follows the posterior-mean convention in archived version 1 of Willard et al., expressed for efficacy maximization¹². Under independent fitted GPs and a fixed incumbent, constrained EI factors into EI times PF^{21–23}. The updated design of Willard et al. instead averages this product over posterior draws of GP parameters and draw-specific constrained optima¹³. Here cEI uses the same fitted Matérn GPs, assignment restrictions, stopping, and final-selection rules as the other acquisitions, so the comparison concerns allocation objectives under our common protocol. It favors combinations with larger expected efficacy improvement and stronger posterior support for meeting the toxicity limit. It does not explicitly calculate how a new cohort would change the final selection.

One-cohort lookahead (cKG). For each eligible dose pair, cKG averages the posterior mean efficacy of the selection that would follow a new cohort, accounting for its possible efficacy and toxicity outcomes. The calculation uses only currently available combinations. Let $\bar{Y}_e(d)$ and $\bar{Y}_g(d)$ be the hypothetical cohort means at d . Averaging over both outcomes gives the score

$$\alpha_{\text{cKG}}(d) = \mathbb{E}\{V_t^+(d) \mid D_t\} - V_t. \quad (4)$$

Each hypothetical cohort updates both GPs with the current hyperparameters held fixed; a superscript $+$ denotes the updated quantities. Let $A_\tau^+ = \{x \in \mathcal{G} : \text{PF}^+(x) > \tau\}$. The updated selection set is $C_t^+ = A_\tau^+ \cap \mathcal{G}_t$ when this intersection is nonempty; otherwise it is the singleton containing the available combination with the largest z_g^+ . We set $V_t^+(d) = \max_{x \in C_t^+} \mu_e^+(x)$ and assign the next cohort to the eligible combination with the largest cKG score. When $A_\tau \cap \mathcal{E}_t$ is nonempty, maximizing cKG maximizes the expected selection value after one cohort under the fitted models. Under the independent Gaussian models, toxicity updates partition the calculation into intervals with a fixed selection set; the efficacy expectation can be integrated analytically within each interval (Appendix A.2.1). Appendix A.2 gives the score decomposition. Later cohorts, future dose availability, stopping, and the hyperparameter refit following an observed cohort are not included in the lookahead.

Targeted mean squared error (tMSE). With equal candidate weights, the zero-bandwidth noisy-level-set form of the criterion introduced by Picheny et al.²⁴ and studied by Lyu et al.²⁵ ranks combinations by

$$\alpha_{\text{tMSE}}(d) = \sigma_g(d) \exp\{-z_g(d)^2/2\} \quad (5)$$

up to a positive constant common to all candidates. tMSE favors combinations with uncertain toxicity and a posterior mean toxicity near the limit, but assignment remains subject to $A_\tau \cap \mathcal{E}_t$. It uses neither efficacy nor the effect of a new cohort on the final dose choice.

Learning toxicity across the grid (entropy). Let $h(p) = -p \log p - (1-p) \log(1-p)$ be binary entropy, with $0 \log 0 = 0$. The entropy score is

$$\alpha_{\text{ent}}(d) = \sum_{x \in \mathcal{G}_t} [h\{\text{PF}(x)\} - \mathbb{E}\{h(\text{PF}^+(x)) \mid D_t\}]. \quad (6)$$

The expectation averages over possible toxicity cohort means $\bar{Y}_g(d)$; PF^+ uses each resulting update, with fitted parameters and $\mathcal{G}_t$ held fixed. The score sums the expected reduction in uncertainty about whether each available combination meets the mean-toxicity limit. Uncertainty is calculated separately for each combination, rather than for the acceptable set as a whole. The entropy-based design of Domenicano et al.¹⁰ learns which dose is optimal. Here, we learn only toxicity, drawing on methods that identify where a response surface falls below a threshold^{26–28}. Numerical details and checks are in Appendix A.3.

2.3 Trial designs and outcome measures

We compare full-grid and gradual dose availability. In the full-grid design, each stratum contributes 40 response records: four initialization observations generated at stratified continuous dose locations, followed by 18 two-participant cohorts assigned on $\mathcal{G}$. There is no early stopping. Allocation summaries use the 36 post-initialization assignments. The four initial observations are model inputs; obtaining them in a clinical trial would require a separate initialization protocol.

In the gradual-availability analysis, enrollment is capped at 40 participants per stratum. The first cohort of two participants starts at $(0, 0)$, after which a fixed schedule makes more combinations available. These initial on-grid assignments count toward enrollment and allocation outcomes. The schedule is not response-guided escalation. Before the full grid is available, previously assigned combinations are excluded while any available unassigned combination remains. This restriction can prevent returning to an observed, tolerated combination. The assignment fallback is used on the first two consecutive occasions when no permitted combination passes the criterion; a third consecutive occasion stops the trial without selection. At maximum enrollment, final selection uses the full grid. Because availability, eligibility, stopping, and enrollment change together, this analysis cannot isolate the effect of escalation. Appendix A gives the schedule.

We report true mean efficacy at final selection and how often that selection exceeds the true mean-toxicity limit, retaining fallback selections. For gradual availability, efficacy is summarized only among trials that make a selection. The above-limit selection percentage is reported both among all initiated trials and among selecting trials, alongside the stopping rate. Stopped trials contribute no above-limit selection to the all-trial percentage; no efficacy value is assigned to them.

The main allocation outcome is the percentage of the 36 post-initialization participants assigned above the limit in full-grid runs. Under gradual availability, where enrollment varies, we report the mean number N_{above} assigned above the limit per initiated trial, including the two initial participants, together with mean enrollment. Both outcomes concern assignments to combinations with above-limit true mean toxicity, not observed adverse events. Appendix A.7 gives the formal definitions.

2.4 Simulation scenarios

We use four synthetic scenarios: OSA-derived¹², logistic, MARIPOSA-motivated¹⁸, and Gaussian bump. The Gaussian-bump scenario has nonmonotone mean toxicity with substantial changes between adjacent grid points (Appendix C). Let M denote the number of independent simulation replicates. The first and last scenarios use $M = 200$; the others use $M = 100$. Within a replicate,

rules share random-number streams for initialization and outcome errors, although adaptive assignments can diverge. Each scenario is evaluated in two strata and at $\tau \in \{0.5, 0.6, 0.7, 0.8, 0.9\}$, with 0.7 as the reference. Gradual availability is evaluated for OSA at 0.7 and 0.9. Strata have separate simulated trials; shared accrual and cross-stratum borrowing are not evaluated. Efficacy and toxicity outcomes are continuous with Gaussian errors. Both outcomes from each cohort are assumed available before the next assignment; accrual time, delayed outcomes, and missingness are not simulated.

For OSA, efficacy is simulated AHI4 reduction in events/hour on the published synthetic response surface¹²; larger values represent greater reduction. The OSA strata represent mild/moderate ($z = 0$) and severe ($z = 1$) disease; toxicity uses the synthetic natural-log scale $\log(B + 0.5)$, where B is a weighted adverse-event burden score, with mean-score limits of 1.5 and 2.0, respectively. The other three scenarios use their own synthetic efficacy scales. Differences between percentages are reported in percentage points (pp).

We summarize paired replicate-level contrasts by their mean, Monte Carlo standard error (MCSE), and two-sided 95% t interval. Each surface-level summary weights two strata and five τ settings equally within replicate. We report surfaces separately; each average describes the settings evaluated on that surface, rather than a clinical population or a single trial design. The main comparisons are exploratory, with pointwise, unadjusted intervals. Monte Carlo intervals quantify simulation error under the specified data-generating models, not uncertainty about clinical treatment effects.

2.5 Design of the sensitivity analysis

A nested OSA analysis compares all four rules at $\tau = 0.7, 0.9$ without early stopping. Each path is analyzed after 20, 40, and 80 total response records, corresponding to 16, 36, and 76 post-initialization assignments. The analysis model assumes a toxicity-outcome SD equal to half, the same as, or twice the data-generating SD, denoted $c_\sigma \in \{0.5, 1, 2\}$. The true surfaces, true outcome SDs, and efficacy model are unchanged. Allocation is summarized as a percentage of post-initialization assignments. Simultaneous 95% max- t Monte Carlo intervals cover the 54 displayed contrasts. This analysis is limited to the assumed toxicity-outcome SD; it does not calibrate clinical enrollment or assess other model errors.

Replication followed a prespecified Monte Carlo precision rule; Appendix B gives the rule and its cap^{29–31}.

2.6 Software and computation

In a fixed-data CPU benchmark of the installed pre-rename software candidate 0.2.1rc4, cKG scoring on the 25-point dose grid took 0.105–0.114 seconds; fitting both response GPs took 0.136–0.147 seconds. The benchmark used three fixed OSA datasets with 4, 20, and 40 observations, rather than adaptive trial paths (Appendix A.4).

3 Results

3.1 Efficacy and toxicity in the OSA-derived scenario

Table 1 summarizes efficacy and toxicity for all four rules. Relative to cEI, cKG increased final-selection efficacy by 0.739 events/hour (95% Monte Carlo interval, 0.504 to 0.974), above-limit final selection by 8.30 pp (5.74 to 10.86), and above-limit allocation by 2.82 pp (1.53 to 4.11). The

efficacy gain came with more above-limit assignments and final selections, so it did not establish an overall benefit–risk advantage. This summary weights five cutoff settings equally. Table 2 reports all three outcomes separately at each cutoff. Appendix A.11 reports acceptable selection within an efficacy margin of the grid optimum.

Table 1: Absolute operating characteristics in the primary OSA-derived analysis under full-grid availability.

Rule	AHI4 reduction at final selection (events/hour)	Above-limit final selection (%)	Above-limit allocation (%)
cKG	7.46 (0.15)	18.80 (1.19)	17.92 (0.61)
tMSE	7.36 (0.14)	11.85 (1.09)	27.08 (0.79)
Entropy rule	8.02 (0.15)	18.65 (1.50)	22.42 (0.78)
cEI	6.72 (0.15)	10.50 (1.14)	15.10 (0.84)

Note. Entries are descriptive means over $M = 200$ replicates, weighting the two strata and five τ values equally. Parentheses give Monte Carlo standard errors (MCSEs), not between-trial standard deviations. Above limit refers to true mean toxicity above the prespecified limit. Allocation uses the 36 post-initialization participants and excludes the four initialization records. All trials made a final selection (100%), including prespecified fallback selections. Efficacy is simulated AHI4 reduction in events/hour. Table 3 contrasts were calculated before rounding.

Table 2: Joint efficacy–toxicity contrasts for cKG versus cEI in the OSA-derived scenario.

τ	Final-selection efficacy difference	Above-limit final selection (pp)	Above-limit allocation (pp)
0.5	1.19 [0.76, 1.63]	13.00 [6.82, 19.18]	6.50 [4.38, 8.62]
0.6	1.03 [0.62, 1.45]	10.50 [5.43, 15.57]	4.07 [2.20, 5.94]
0.7	0.91 [0.51, 1.31]	9.75 [5.18, 14.32]	2.33 [0.69, 3.98]
0.8	0.39 [0.04, 0.74]	5.50 [1.95, 9.05]	1.33 [0.01, 2.65]
0.9	0.17 [−0.13, 0.47]	2.75 [0.12, 5.38]	−0.14 [−1.33, 1.06]

Note. Entries are cKG-minus-cEI differences [pointwise unadjusted 95% Monte Carlo intervals], averaging the two strata within each of $M = 200$ paired replicates. Positive efficacy differences favor cKG; positive differences in above-limit selection or allocation favor cEI. Efficacy differences are in events/hour. Allocation uses 36 post-initialization assignments. Final-selection results retain fallback selections.

In OSA, lower above-limit allocation did not imply fewer above-limit final selections. Relative to tMSE, cKG reduced above-limit allocation by 9.15 pp (95% Monte Carlo interval, 8.12 to 10.19), but increased above-limit final selection by 6.95 pp (4.70 to 9.20). Relative to entropy, the allocation difference favored cKG by 4.50 pp (3.24 to 5.76); the final-selection difference was 0.15 pp (−2.48 to 2.78). That interval does not establish equivalence. Entropy also had higher mean efficacy at final selection than cKG, 8.02 versus 7.46 events/hour (Table 1). Table 3 and Figure 1 give the paired contrasts.

3.2 Efficacy of the final selection: cKG versus cEI

cKG had higher estimated mean efficacy at final selection than cEI at every cutoff on all four surfaces (Figure 2; Appendix Table 8). Pointwise Monte Carlo intervals excluded zero at $\tau = 0.5, 0.6, 0.7$ on every surface. Differences were smaller at 0.8 and 0.9; some intervals then included zero. Efficacy is not pooled across surfaces because its scale differs. In the Gaussian-bump scenario, fallback occurred in 17.25–41.75% of cKG trials and 17.25–41.25% of cEI trials, depending on τ . The pooled cKG–cEI efficacy difference was 0.19 overall, compared with 0.24 among criterion-passing selections and 0.06 among fallback selections (Appendix A.10). These subgroup comparisons involve different trials under the two rules and do not show how either rule would perform without fallback.

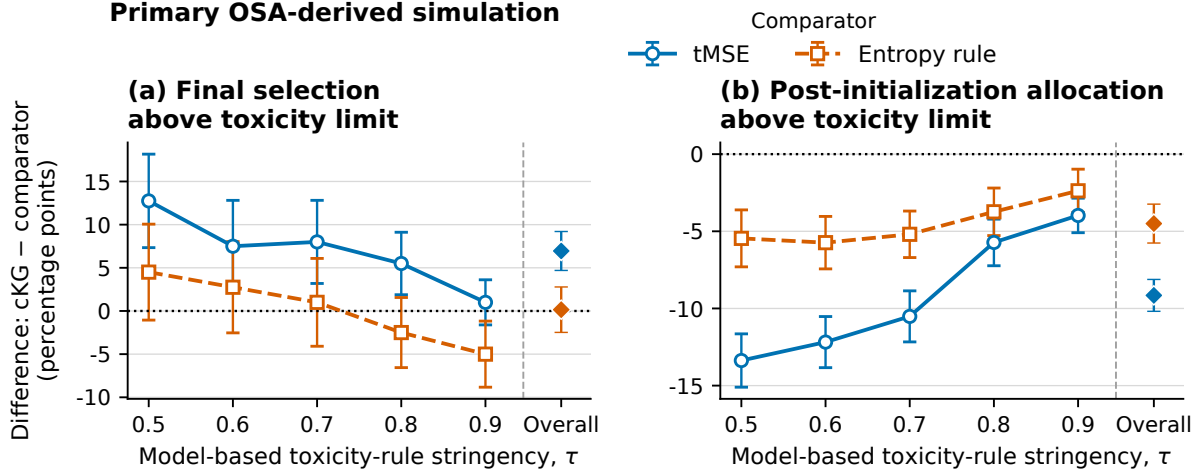


Figure 1: cKG-minus-comparator differences in the primary OSA-derived simulation ($M = 200$): (a) above-limit final selection and (b) above-limit post-initialization allocation. Points average the two strata; “Overall” is a descriptive average over the five τ values, not a single trial design. Bars are pointwise unadjusted 95% Monte Carlo intervals.

Fallback accounted for little of the excess above-limit final selections under cKG in OSA (Appendix Table 9). Criterion-passing selections were truly above limit in 18.55% of all cKG trials and 10.15% of cEI trials; above-limit fallback selections accounted for only 0.25% and 0.35%. The probability of a selection that both passed the criterion and was truly acceptable was 80.55% under cKG versus 88.30% under cEI, a difference of -7.75 pp (95% Monte Carlo interval, -10.38 to -5.12). On the MARIPOSA-motivated surface neither policy used a terminal fallback, yet their above-limit selection rates were 20.40% and 0.20%. Passing the posterior criterion therefore did not ensure that true mean toxicity was below the limit.

3.3 Patterns across synthetic response surfaces

In each surface-level summary, cKG made more above-limit final selections than tMSE, by 4.25–12.10 pp, while allocating less often above the limit, by 2.36–11.37 pp. Pointwise Monte Carlo intervals excluded zero for both outcomes in all four summaries. The Gaussian-bump scenario has a different, nonmonotone geometry (Appendix Figure 5), but several features change together; these comparisons do not identify which feature explains the pattern.

Comparisons with entropy were less consistent. On the logistic surface, cKG both selected above-limit combinations more often and allocated above the limit less often, with intervals excluding zero. On the OSA-derived and MARIPOSA-motivated surfaces, only the allocation intervals excluded zero; on the Gaussian-bump surface, neither interval excluded zero. Appendix Figure 3 shows the complete surface-by- τ profiles.

The efficacy comparisons also varied by surface. Final-selection efficacy favored cKG over tMSE on the logistic and Gaussian-bump surfaces. Compared with entropy, efficacy favored cKG on the logistic surface and entropy on OSA; the remaining pointwise intervals included zero (Appendix Table 7).

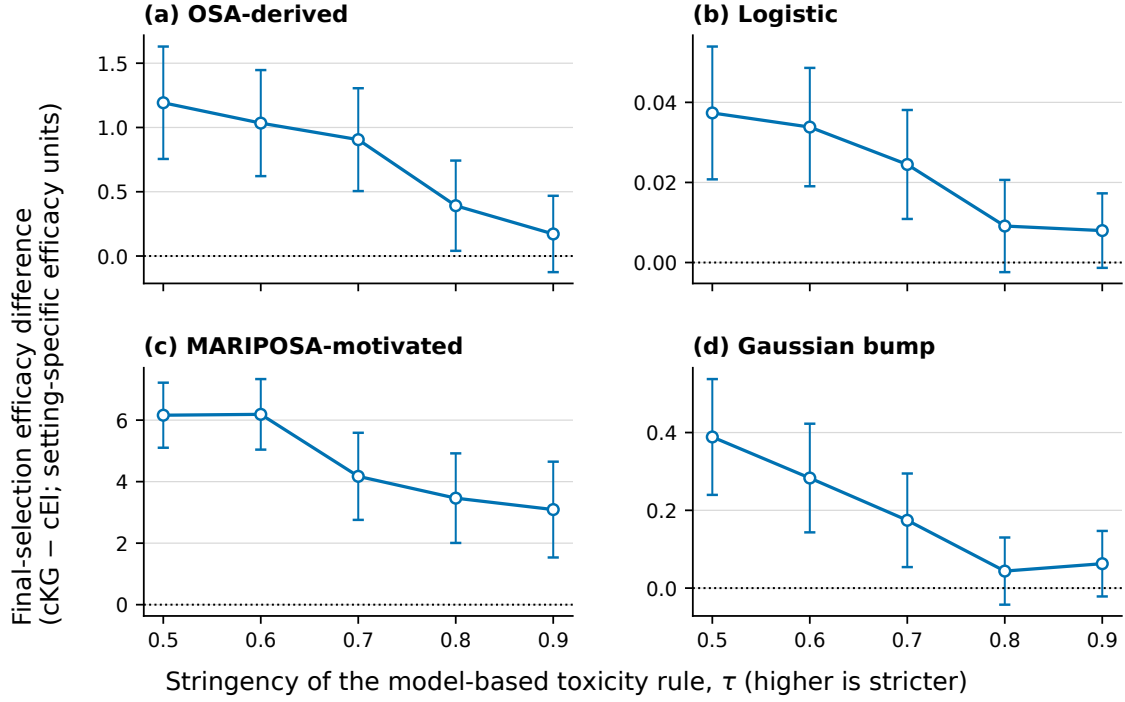


Figure 2: cKG-minus-cEI differences in true mean efficacy at final selection under full-grid availability. Points average the two strata at each τ ; bars are pointwise unadjusted 95% Monte Carlo intervals. Positive values indicate higher true mean efficacy under cKG. Efficacy scales differ by setting. OSA-derived and Gaussian-bump settings use $M = 200$ replicates; the others use $M = 100$.

Table 3: cKG-minus-comparator differences in above-limit final selection and participant allocation by simulation scenario under full-grid availability.

Setting	Comparator	Above-limit final selection (pp)	Above-limit allocation (pp)
OSA-derived	tMSE	6.95 (1.14) [4.70, 9.20]	-9.15 (0.52) [-10.19, -8.12]
OSA-derived	Entropy rule	0.15 (1.33) [-2.48, 2.78]	-4.50 (0.64) [-5.76, -3.24]
Logistic	tMSE	12.10 (1.68) [8.76, 15.44]	-5.32 (0.60) [-6.52, -4.13]
Logistic	Entropy rule	11.90 (1.76) [8.40, 15.40]	-5.07 (0.61) [-6.27, -3.86]
MARIPOSA-motivated	tMSE	5.40 (2.68) [0.08, 10.72]	-11.37 (0.99) [-13.32, -9.41]
MARIPOSA-motivated	Entropy rule	3.60 (2.92) [-2.19, 9.39]	-7.78 (0.98) [-9.72, -5.83]
Gaussian bump	tMSE	4.25 (1.06) [2.16, 6.34]	-2.36 (0.58) [-3.50, -1.22]
Gaussian bump	Entropy rule	1.60 (1.23) [-0.82, 4.02]	-0.10 (0.74) [-1.57, 1.36]

Note. Entries are differences in pp shown as estimate (MCSE) [pointwise unadjusted 95% Monte Carlo interval]. Above limit refers to true mean toxicity above the prespecified limit. Allocation uses the 36 post-initialization participants and excludes the four initialization records. Within each scenario, the two strata and five τ values are weighted equally; these averages are descriptive. OSA-derived and Gaussian-bump rows use $M = 200$ replicates; logistic and MARIPOSA-motivated rows use $M = 100$. Positive selection differences and negative allocation differences mean more above-limit selections and fewer above-limit assignments under cKG, respectively. Final-selection results include prespecified fallback selections.

3.4 Gradual dose availability and stopping

Table 4 reports gradual-availability results for OSA. At $\tau = 0.7$, cKG made above-limit final selections 8.75 pp more often than tMSE (95% Monte Carlo interval, 3.89 to 13.61), while reducing mean N_{above} by 4.50 participants per initiated trial, including stopped trials (3.97 to 5.03 fewer). At $\tau = 0.9$, the corresponding differences were 2.50 pp (0.32 to 4.68) and 1.81 fewer participants (1.42 to 2.19 fewer). cKG also had lower mean N_{above} than entropy at both $\tau = 0.7$ and 0.9, with intervals excluding zero. These are counts per initiated trial, not assignment probabilities: cKG stopped more often than the other rules and consequently enrolled fewer participants. Among trials that made a final selection, cKG had lower mean efficacy than tMSE and entropy at both cutoffs, and higher mean efficacy than cEI; these conditional summaries involve different subsets of trials.

Table 4: OSA-derived operating characteristics under gradual dose availability.

τ	Rule	Above-limit final selection (%)				Mean enrollment	Mean N_{above}
		Final-selection efficacy	All initiated trials	With selection (n_{sel})	Stopped (%)		
0.7	cKG	7.26	18.50	19.07 (388)	3.00	39.14	6.47
0.7	tMSE	7.70	9.75	9.95 (392)	2.00	39.42	10.97
0.7	Entropy rule	8.10	17.50	17.54 (399)	0.25	39.92	10.07
0.7	cEI	6.40	10.25	10.35 (396)	1.00	39.71	3.81
0.9	cKG	5.42	4.50	5.41 (333)	16.75	35.20	4.44
0.9	tMSE	5.70	2.00	2.29 (350)	12.50	36.45	6.24
0.9	Entropy rule	6.07	7.50	8.11 (370)	7.50	37.86	5.78
0.9	cEI	5.19	2.75	3.00 (367)	8.25	37.63	3.28

Note. Each rule- τ row represents 400 initiated stratum-specific trials ($M = 200$ replicates in two strata); larger τ denotes a more stringent toxicity rule. Above limit refers to true mean toxicity above the prespecified limit. Stopped trials contribute zero to the all-trial percentage; parentheses in the conditional column give n_{sel} . Efficacy is mean simulated AHI4 reduction (events/hour) among the n_{sel} trials that made a selection; no efficacy value is assigned to stopped trials. The set of trials contributing to each efficacy mean differs between rules. Mean enrollment and mean N_{above} use all initiated trials. N_{above} includes the two initial on-grid assignments and counts assigned participants, not observed toxicity events.

3.5 Sensitivity to enrollment and assumed toxicity variability

The nested analysis used $M = 500$ replicates, evaluated after 20, 40, or 80 response records with assumed-to-generated toxicity-SD ratios $c_\sigma \in \{0.5, 1, 2\}$. Of the 54 toxicity contrasts, 51 met the prespecified precision criterion. Appendix B describes the procedure and exceptions.

Compared with tMSE, cKG reduced above-limit allocation by 3.08–8.65 pp across the snapshots and assumed SDs; all simultaneous intervals excluded zero. Above-limit final selection was 1.90–9.30 pp more frequent, with intervals excluding zero except at the earliest snapshot when toxicity variability was understated. Neither contrast followed a common monotone pattern across the three SD assumptions.

Results against entropy changed with the assumed toxicity SD. When the SD was understated, cKG allocated less often above the limit at all snapshots and selected above-limit combinations more often at the last two. When the SD was overstated, the allocation contrast reversed at the first two snapshots, and cKG selected above-limit combinations less often at the last two. Under the matching SD, cKG allocated above the limit more often at the first snapshot and less often thereafter; the final-selection interval excluded zero only at the last snapshot. Compared with cEI, cKG made above-limit final selections more often in every setting, with simultaneous intervals excluding zero. The ranking depended on the assumed toxicity SD. Across the nine SD-by-snapshot summaries, cKG had higher mean final-selection efficacy than cEI and lower mean efficacy than entropy; its efficacy relative to tMSE depended on the setting (Appendix Table 11). These efficacy summaries are descriptive and separate from the 54 toxicity contrasts.

4 Discussion

The deterministic evaluator calculates a one-cohort cKG score under independent Gaussian response models, with fitted hyperparameters held fixed during each hypothetical update. In the full-grid simulations, cKG had higher estimated mean final-selection efficacy than cEI at every cutoff on all four surfaces, with smaller differences under stricter criteria. Averaged over the OSA strata and cutoffs, this gain accompanied more above-limit assignments and final selections. Compared with tMSE, cKG assigned fewer participants to above-limit combinations but selected them more often at trial completion. These outcomes describe distinct trade-offs rather than a single ranking of the rules.

Entropy had higher final-selection efficacy than cKG in the primary OSA summary. Although it does not target efficacy directly, reducing uncertainty about toxicity may allow more high-efficacy combinations to qualify for final selection. cKG also learns toxicity, but assigns the next cohort according to its expected effect on selection value under the current models. Computing that score exactly does not guarantee better performance over the whole trial.

For trial statisticians, the starting point is the final decision: the toxicity criterion, any minimum efficacy requirement, the evidence needed to recommend a combination, and the option to recommend none. Within a finite-grid BO design using the independent fitted GPs and recommendation rule studied here, cKG can replace cEI for cohort allocation while eligibility, stopping, and final selection remain fixed. The complete protocol should then be evaluated, reporting efficacy together with above-limit allocation, final-selection toxicity, fallback use, and stopping. The `dose-combination-bo` package supports these comparisons. Our cEI shares the acquisition form used by Willard et al., but the comparison does not reproduce their complete design^{12;13}. In particular, their updated fully Bayesian fitting and optimal-dose entropy stopping differ from our fitted-model comparison and toxicity-classification entropy acquisition.

Several features limit clinical interpretation. Posterior probability cutoffs were not clinically calibrated. The selection rule imposes no additional toxicity penalty below the limit and can recommend an untried combination or a fallback that fails the criterion. Gaussian-bump results include many fallbacks; OSA and MARIPOSA-motivated results also show that criterion-passing selections can exceed the true mean-toxicity limit. The primary design uses four initialization observations, while gradual availability starts on-grid and follows a fixed opening schedule. Both assume that cohort outcomes are immediately available. Neither addresses delayed outcomes, dropout, shared accrual, permanent dose elimination, or independent safety monitoring. Every scenario contains an acceptable grid dose; no minimum clinically important efficacy or all-inactive scenario is specified. Independent GPs omit efficacy–toxicity dependence, and plug-in fitting omits hyperparameter uncertainty. The primary simulations also use the data-generating outcome variances. The sensitivity analysis varies only the assumed toxicity–outcome SD, not kernel choice, dose scaling, or prior-data conflict. A clinical application would require justified outcome definitions, observation windows and toxicity limits, separate handling of serious adverse events and discontinuation, and case-specific validation of the complete decision rules. Published arm summaries cannot supply the required individual-level outcome distributions.

5 Conclusions

The proposed cKG rule calculates how one additional cohort would change the value of the final dose-selection rule under the fitted response models. Under full-grid availability, cKG had higher estimated mean final-selection efficacy than cEI at every tested cutoff on all four surfaces. Pointwise Monte Carlo intervals excluded zero at $\tau = 0.5, 0.6, 0.7$, but not in every stricter-cutoff comparison. Averaged over the OSA strata and cutoffs, cKG made more above-limit assignments and final selections than cEI, and fewer above-limit assignments but more above-limit final selections than tMSE. Allocation and final selection should therefore be evaluated separately, with efficacy reported alongside toxicity, fallback use, and stopping. The accompanying software provides these allocation objectives for evaluation under explicitly specified toxicity criteria and final-selection rules.

Conflict of interest

The authors declare no conflict of interest.

Funding

No specific funding was received for this work.

Ethics approval

Not applicable; this study used simulated data and published arm-level summaries only.

Patient consent

Not applicable.

Clinical trial registration

Not applicable.

Data availability statement

No patient-level clinical data were used. The maintained software, frozen controller reference, selected simulation records, analysis scripts, and summary outputs are held at <https://github.com/clairezhuwang/dose-combination-bo>. The included records support recalculation of full-grid and gradual-availability summaries and the post hoc selection-margin analysis. The complete original study archive, including the enrollment-noise path shards required to replay calibration snapshots, is retained separately and is not included in this repository. Public access to that complete archive and a persistent archive identifier remain to be established for this candidate.

Use of generative AI

Generative AI tools, including OpenAI Codex, assisted language editing, code review, preparation and execution of verification and post-processing scripts, and software packaging. AI-assisted verification and post-processing scripts were run separately from the original simulation experiments. The authors remain responsible for reviewing the generated material, analysis, interpretation, references, and final submitted text.

Author contributions

X.W. conceived the study, adapted the method, developed its evaluator, implemented and ran the simulations, and drafted the manuscript. T.L. supervised the work and reviewed and edited the manuscript.

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